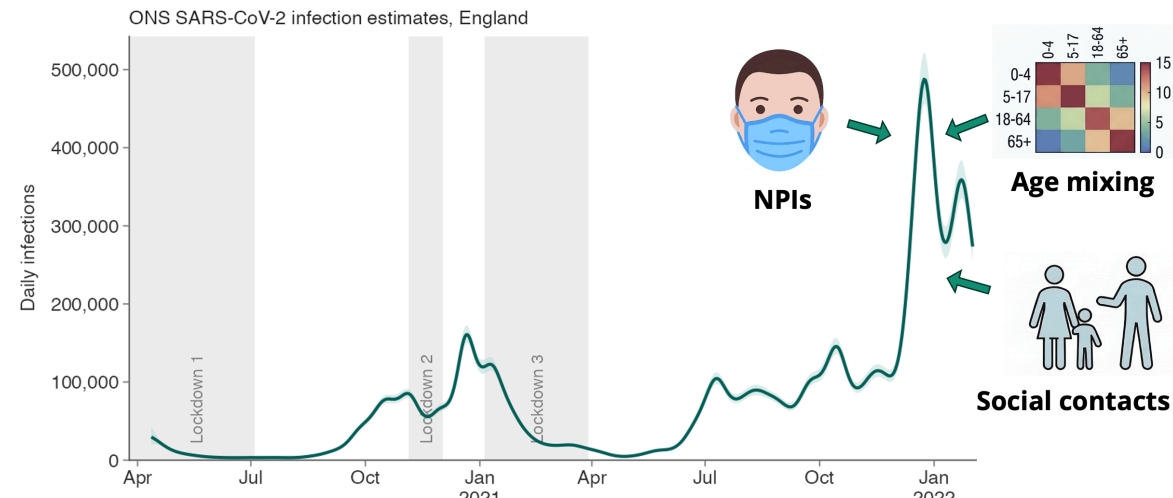


'Understanding and predicting behavioural drivers of epidemic dynamics'



A bit about me



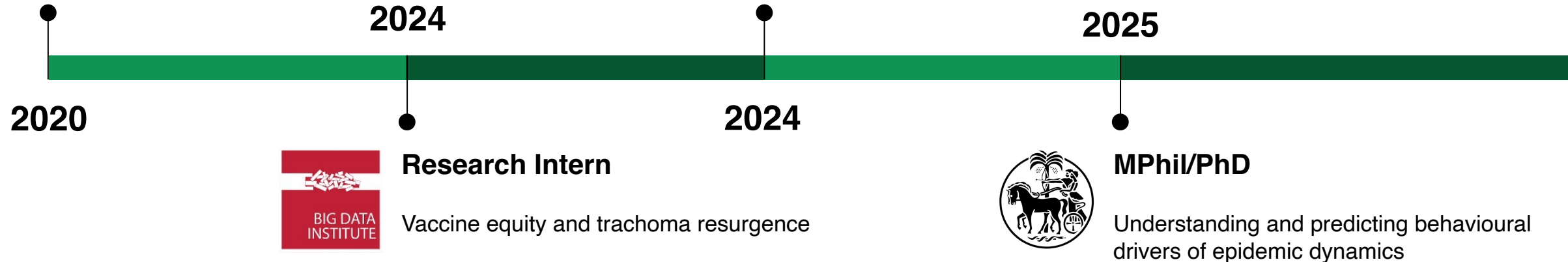
MMath, Mathematics

Branching process models for estimating disease outbreak probabilities



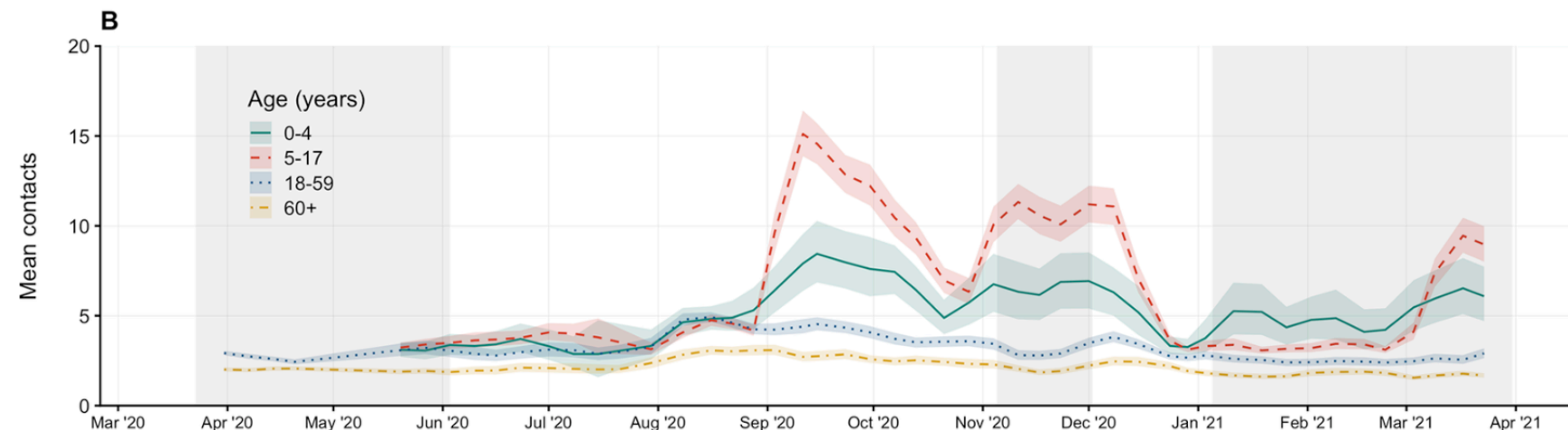
MSc, Health Data Science

Using machine learning to understand vaccine responses to influenza



Respiratory transmission is mediated by behaviour

- Transmission needs **susceptible** host, pathogen and contact - NPIs modify risk
- Some 'fixed' behaviour - age, household, occupation
- The rest is variable - policy, risk and context
- The modelling choice: **which behavioural signal to use**



Contacts, mobility and attitudes capture different aspects

Contacts (CoMix)

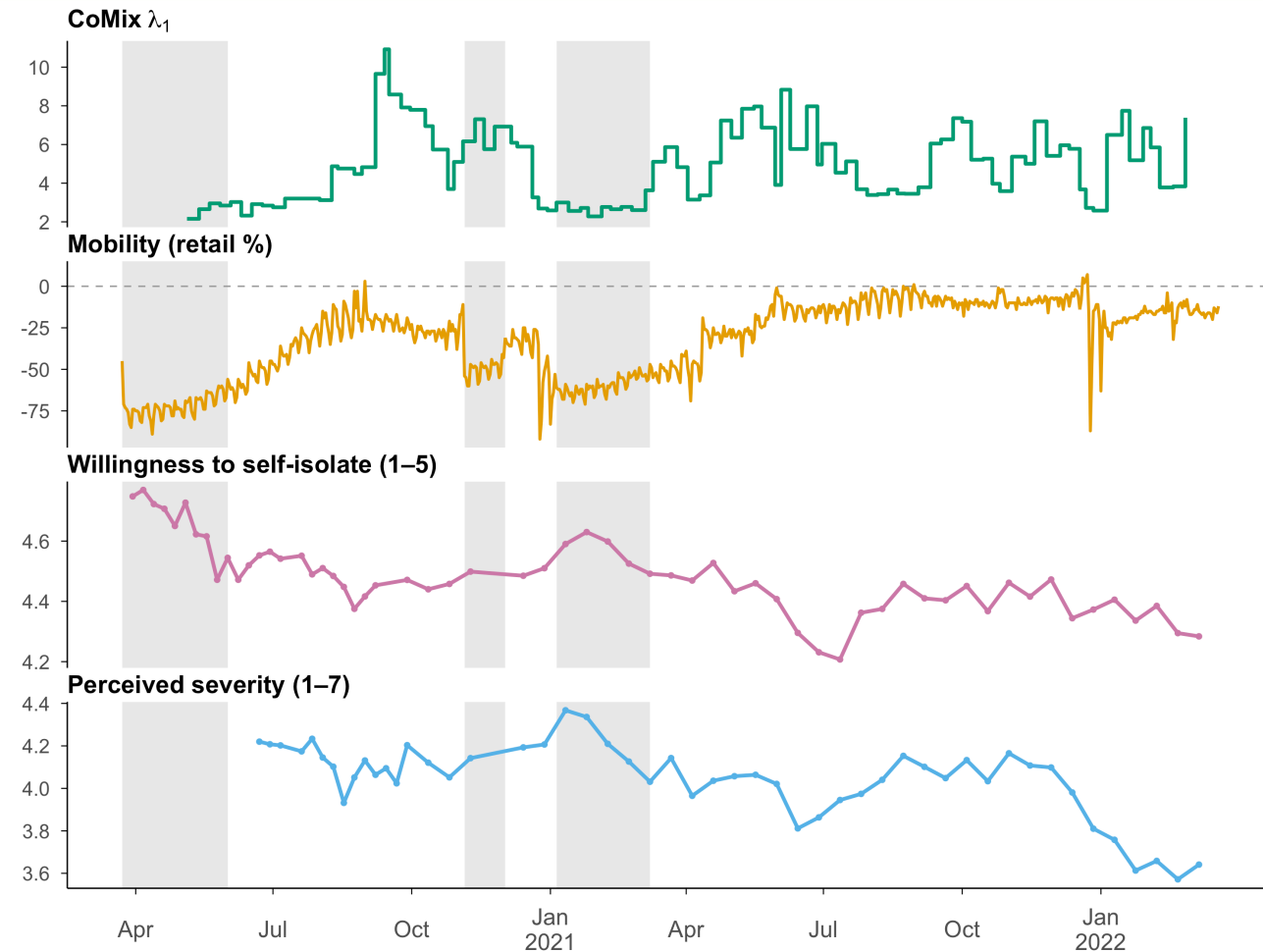
age-structured weekly contact counts

Mobility (Google)

daily relative time spent in settings

Attitudes (CoMix, YouGov)

risk perception, trust, self-efficacy



Streams differ - overlap, cadence, coverage, cost, lag, coarseness...
Can we **combine** to explain and forecast transmission?

Aim 1: Compare behavioural streams in renewal framework

$$I_t \sim \text{Poisson}(R_t \Lambda_t) \quad \text{where} \quad \Lambda_t = \sum_{s=1}^{s_{\max}} I_{t-s} w_s$$

total infectious pressure generation interval

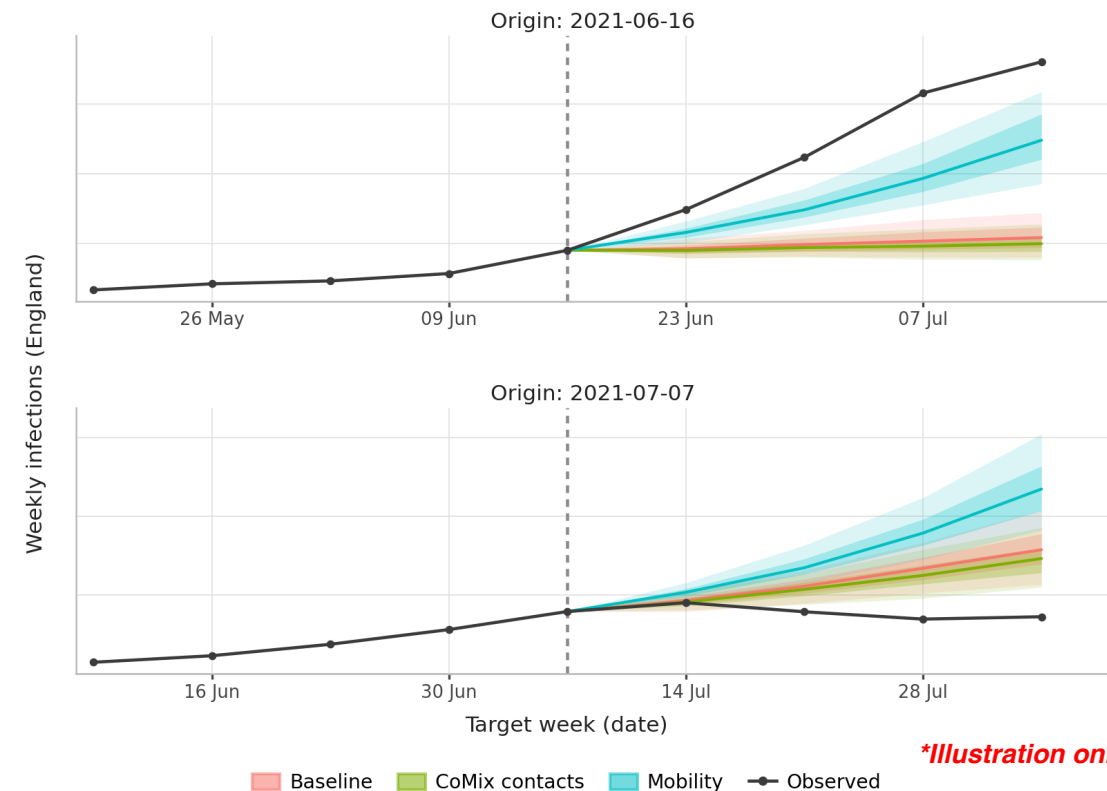
$$\log(E[I_t]) = \log(R_t) + \log(\Lambda_t)$$

$$\log(R_t) = \alpha_0 + s(t) + \sum_{k=1}^K \beta_k X_{k,t-\ell_k}$$

behaviour enters here

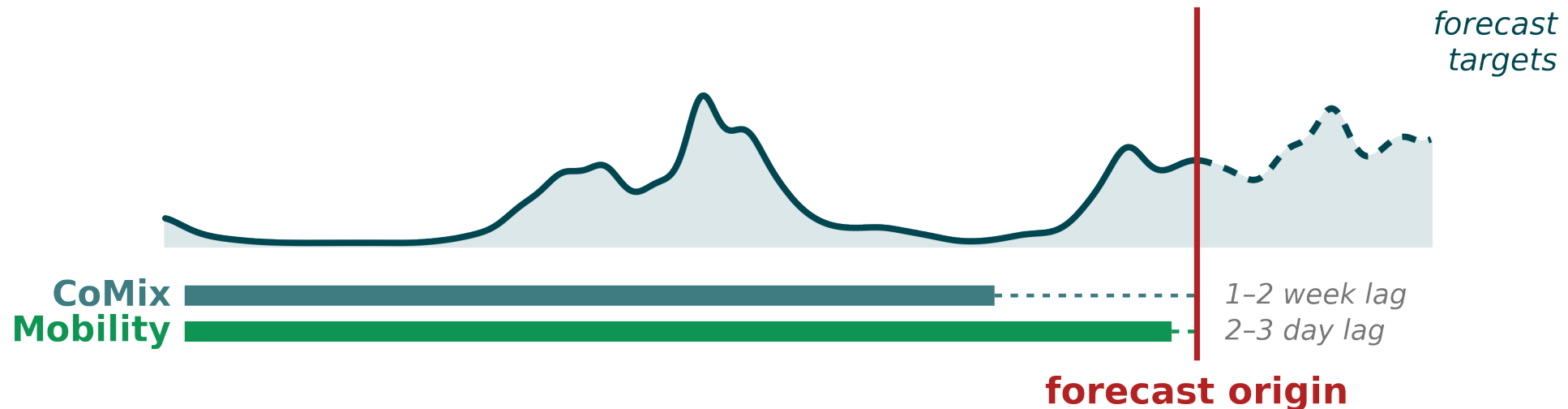
- Infections: ONS CIS via {inc2prev}
- Scored **out of sample** against baseline

Model	Covariates
Baseline	past incidence only
Mobility	+ Google Mobility
CoMix	+ NGM dominant eigenvalue λ_t
Combined	+ CoMix and Mobility



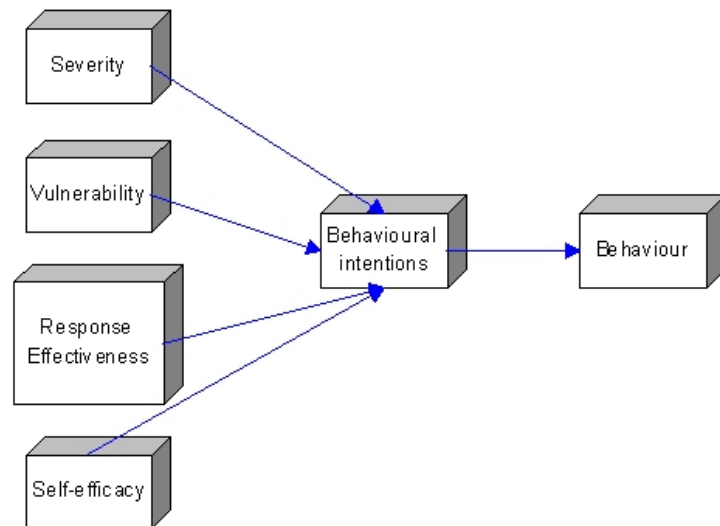
Aim 2: How about real-life data delays?

- **Aim 1** used finalised data = optimistic
- Real life has reporting delays and recent infections missing
- Refit using data available **at the time**
- Forecast R_t with **uncertainty**

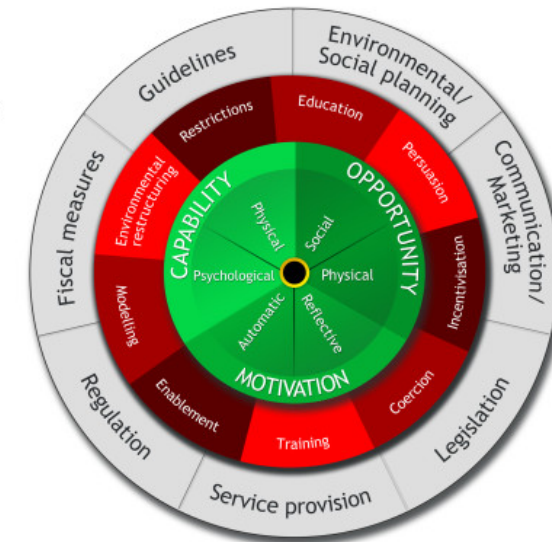
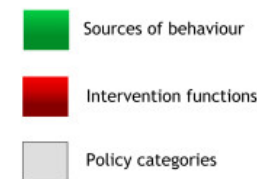


Can we understand why behaviour changed?

- Contacts and mobility record what people **did**, not **why**
- Drivers: perceived risk, trust, efficacy, fatigue
- Frameworks like **PMT** and **COM-B** describe determinants of behaviour change
- Structural **constraints** can override intention



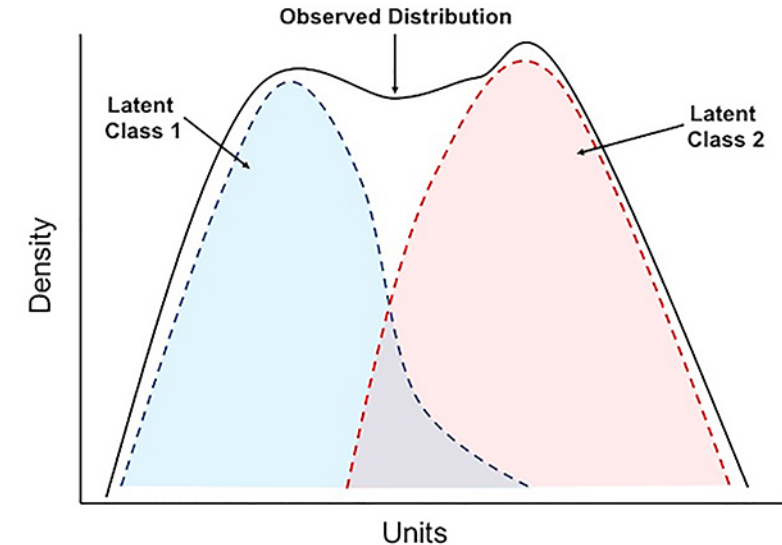
Rogers 1975; Protection Motivation Theory



Michie et al. 2011; COM-B Behaviour Change Wheel

Aim 3: Do meaningful archetypes exist in the population?

- Five **driver indicators** measured every wave in CoMix
- **Latent class analysis** groups people by joint profile
- Archetype → **non-household contacts**



Sinha et al. 2021; Practitioner's Guide to Latent Class Analysis

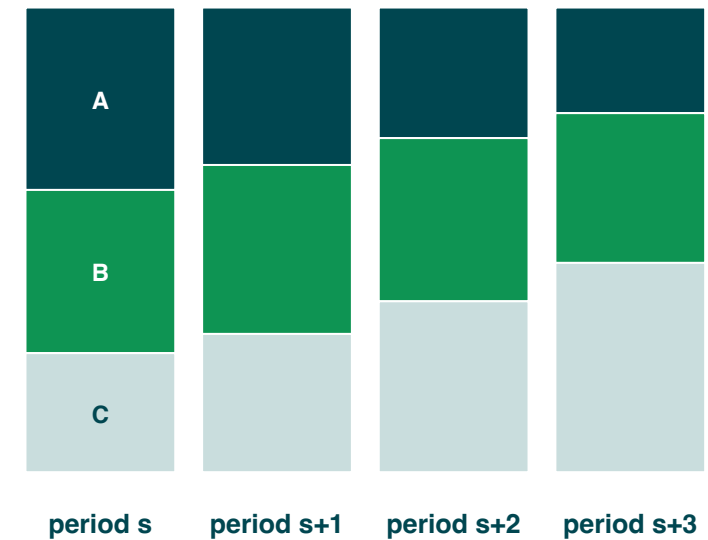


...moderated by constraints and sociodemographics

Mean behaviour overlooks composition

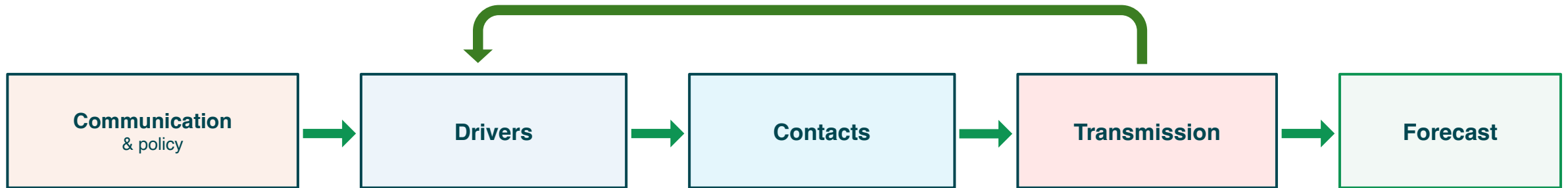
- The same average can hide a very different **mix of people**
- Shifting mixes can **move transmission**
- **Epidemic** and **policy** shifts can move that mix

$$C_t = \int b(x_t, \theta) dF_t(\theta) \approx \sum_{k=1}^K \pi_{k,t} b_k(x_t)$$



Models using **averages** may mispredict transmission when composition shifts

A potential avenue: communication as real-time signal



- A **dated, attributed** record of official communication - daily
- A candidate **real-time signal**, available at the forecast origin
- Does it improve a **nowcast**, beyond mobility and policy?

*Predictive, **not** causal*



Aim 1

ABSTRACTION

Compare contacts and mobility in one renewal framework

Aim 2

DATA NEEDS

Re-do in real time, on the data available then

Aim 3

POLICY RESPONSE

Attitudinal archetypes predicting contacts - who responds, and who cannot

Future avenue?

DIRECTION

Communication as a real-time signal

Working around: different data from different populations with different variables